

Term Project — Use-Case Catalog

Fifteen starting points for a mentor-supervised commercial analytics project

Applied Data Science and AI in Pharma Commercial · Columbia University · Fall 2026

Distributed Week 1 (Sep 14) · teams of 3–4 formed by Sep 21 · mentors introduced Sep 21 · project is 40% of the grade

Deliverables (Dec 14): 20-minute presentation + 5 min Q&A · re-runnable code repository with README · 2–4 page executive memo · mentor acknowledgment

How to use this catalog. The syllabus lists five acceptable archetypes; this catalog expands them to fifteen briefs so your team can find a problem it actually cares about. Each brief names the commercial question, the stakeholder whose behavior you are modeling, a realistic data plan (synthetic data you generate, course datasets, or public sources), the methods the course will cover, what “good” looks like, and the traps mentors will probe. You may also propose your own; the test is the same — would this output be credible inside a real pharma organization?

Ground rules that apply to every project

No real patient-level data. Use synthetic or fully de-identified public data; never paste proprietary or patient-identifiable data into an external AI tool.

Name the stakeholder. Every project must state whose decision it informs (brand lead, sales ops, market access, medical) and whose behavior it models (HCP, payer, patient).

Evaluate honestly. A working prototype with a clear evaluation and limitations section beats a polished demo. Mentors will ask “how do you know it works?”

Disclose AI use per the syllabus policy: tool, prompts, and what you changed.

Scope for 10 weeks of part-time work. Most strong projects answer one question well.

1. HCP segmentation + Next Best Action engine

Track: Data science (syllabus archetype) · Course weeks: 4, 5, 3

Commercial question	Which HCPs should GLUCOREN's field and digital channels prioritize, through which channel, how often — and how would we know it worked?
Decision owner / behavior modeled	Sales operations and marketing · HCP prescribing and channel response
Data plan	Generate a synthetic HCP panel (5–10k NPIs) with specialty, decile, patient mix, monthly NRx/TRx, and channel touch history (calls, emails, web); or extend the Week 2/3 course datasets.
Methods	Feature engineering from Rx/claims (Wk 4); clustering and supervised segmentation; uplift or affinity modeling; a scoring layer with frequency caps and channel substitution rules (Wk 5); a simulated A/B evaluation.
What good looks like	Segments a marketer can name and act on; an NBA table per HCP with a rationale; an honest estimate of incremental TRx vs. a “business as usual” policy, with uncertainty.
Traps mentors will probe	Segment stability over time; leakage from using future Rx in features; fatigue — does your policy contact some HCPs 40 times a quarter?; interpretability for a brand team.

2. Marketing mix model calibrated to a launch curve

Track: Data science (syllabus archetype) · Course weeks: 3, 6

Commercial question	What did each promotional channel contribute to a brand's first three years, and how should next year's budget be reallocated?
Decision owner / behavior modeled	Brand lead and finance · aggregate market response to promotion
Data plan	Weekly synthetic brand data: TRx, spend by channel (detailing, samples, digital, conferences, DTC), access events, competitor entry, seasonality; calibrate to a public analog launch curve (e.g., from IQVIA or company 10-Ks).
Methods	Adstock (geometric/Weibull) and Hill saturation; linear and Bayesian MMM (PyMC-Marketing); base/incremental decomposition; ROI curves and a constrained reallocation (Wk 3).
What good looks like	A decomposition a CFO can follow; channel ROI with credible intervals; a reallocation with an explicit statement of what would change the answer.
Traps mentors will probe	Identifiability with collinear spend; ignoring access/competitor covariates; presenting point ROIs without ranges; using invented net revenue per TRx instead of a stated assumption.

3. Patient-finding / patient-journey model

Track: Data science (syllabus archetype) · Course weeks: 2, 5

Commercial question	Which patients in a claims population look like our treated cohort but are undiagnosed or untreated — and where are they (which HCPs, which geographies)?
Decision owner / behavior modeled	Medical and marketing · patient diagnosis and treatment pathways; HCP referral behavior
Data plan	Synthetic longitudinal claims (medical + pharmacy) for ~50k patients with a rare or under-diagnosed condition; the Week 2 DLBCL dataset is a template for structure.
Methods	Cohort definition with ICD-10/NDC/HCPs (Wk 2); sequence features and sequence models or gradient boosting; survival analysis for time-to-treatment; calibration and precision at top-k.
What good looks like	A ranked list of HCPs with the most likely undiagnosed patients, a validation on held-out diagnosed patients, and a discussion of how the output would be used compliantly (no patient targeting).
Traps mentors will probe	Label leakage from post-diagnosis codes; base-rate confusion; privacy — the output must be HCP-level, not patient-level; overfitting to synthetic-data artifacts.

4. Adherence and persistence prediction with intervention design

Track: Data science · Course weeks: 2, 5

Commercial question	Which newly started patients are likely to discontinue by day 90, and which support interventions (copay, nurse call, reminders) would change that?
Decision owner / behavior modeled	Patient services and brand · patient refill behavior
Data plan	Synthetic pharmacy claims with fills, days' supply, copay, pharmacy type, plus a simulated patient-support-program enrollment table.
Methods	PDC/MPR computation (Wk 2); classification with calibrated probabilities; survival models; uplift modeling or a simulated experiment for the intervention.
What good looks like	A persistence curve by segment, a risk score at first fill with its lift chart, and a recommended intervention policy with expected retained TRx.
Traps mentors will probe	Immortal-time bias; defining "discontinuation" without a gap rule; assuming correlation (copay ↔ drop-off) is causal.

5. Payer access impact analysis

Track: Data science · Course weeks: 1, 2, 6

Commercial question	What did a formulary change (tier move, prior-auth added, exclusion) do to NBRx, abandonment and net price — and what is the value of winning it back?
Decision owner / behavior modeled	Market access · payer coverage decisions and patient/HCP reaction
Data plan	Synthetic claims with plan IDs and payer type, with formulary events injected on known dates; public formulary documents for realism.
Methods	Event study / difference-in-differences; interrupted time series; a gross-to-net calculation linking volume to net revenue (Wk 1 exercise).
What good looks like	A clean before/after estimate with a control group, translated into \$ using a stated net price, and a one-page contracting recommendation.
Traps mentors will probe	Confounding by simultaneous promotion; spillover between plans; ignoring that rebates change the net price at the same time as volume.

6. Launch forecast with analogs and an epidemiology funnel

Track: Data science · Course weeks: 6, 1

Commercial question	What will a new brand's first three years of TRx look like, with a defensible range, and what drives the width of that range?
Decision owner / behavior modeled	Brand and finance · patient flow through diagnosis and treatment; HCP adoption
Data plan	Public epidemiology (prevalence, diagnosis and treatment rates), public analog launch curves, synthetic access assumptions.
Methods	Epi funnel (incidence → diagnosed → treated → eligible → share); Bass diffusion; analog selection and scaling; scenario and Monte Carlo ranges; hierarchical reconciliation (Wk 6).
What good looks like	A forecast with explicit assumptions and a tornado chart of sensitivities; a comparison with a naïve analog average; a one-page “what would change our mind.”
Traps mentors will probe	False precision; choosing analogs by therapeutic area only; forgetting access ramp and inventory pipeline fill.

7. Loss-of-exclusivity erosion model

Track: Data science · Course weeks: 6, 1

Commercial question	How fast will a brand's volume and net price erode after generic entry, and when should promotion stop?
Decision owner / behavior modeled	Brand and finance · payer switching and generic substitution behavior
Data plan	Public IQVIA/FDA data on historical LOE events; a synthetic brand P&L to apply the erosion curve to.
Methods	Time-series and scenario analysis; erosion-curve clustering by molecule type, number of generic entrants, and channel mix; break-even of promotional spend against remaining contribution.
What good looks like	An erosion curve library with uncertainty, applied to the P&L to recommend a promotion wind-down date.
Traps mentors will probe	Treating biologics like small molecules; ignoring authorized generics; ignoring that net price erodes faster than volume in some markets.

8. KOL and influence network analysis

Track: Data science · Course weeks: 4, 7

Commercial question	Which physicians shape prescribing in a specialty and region, and how should medical affairs prioritize engagement?
Decision owner / behavior modeled	Medical affairs · peer influence on HCP behavior
Data plan	Public publication and co-authorship data (PubMed), Open Payments, synthetic referral/co-management links from claims.
Methods	Graph construction and centrality; community detection; diffusion of prescribing behavior on the network; a ranking with explanations.
What good looks like	A ranked, explainable KOL map; validation against known guideline authors; a compliant engagement plan (non-promotional).
Traps mentors will probe	Confusing volume with influence; data-source bias (academics over community physicians); compliance boundaries between medical and commercial.

9. Field-note insight extraction

Track: NLP (syllabus archetype) · Course weeks: 7

Commercial question	What are reps and MSLS hearing about access barriers, efficacy perceptions and competitors — by segment and region — and how is it trending?
Decision owner / behavior modeled	Brand and medical leadership · HCP sentiment as reported in the field
Data plan	Generate 5–20k synthetic CRM call notes with an LLM from a specified distribution of themes, then treat the generator's labels as (noisy) ground truth.
Methods	Embeddings and clustering; topic models; LLM-based structured extraction with a schema; sentiment; an evaluation against the labels and a human review sample (Wk 7).
What good looks like	A dashboard-ready table of themes by segment/region/month with representative quotes, precision/recall of the extractor, and a discussion of adverse-event handling.
Traps mentors will probe	Adverse-event mentions must be routed to pharmacovigilance — design for it; synthetic notes being too clean; hallucinated themes.

10. Social-listening and adverse-event triage

Track: NLP · Course weeks: 7

Commercial question	Among posts mentioning a brand, which describe a possible adverse event vs. a complaint, a question, or noise?
Decision owner / behavior modeled	Pharmacovigilance and brand · patient voice
Data plan	Synthetic social posts (LLM-generated with labels) and public de-identified forums where licensing allows.
Methods	Classification with general vs. domain models (BioBERT/ClinicalBERT vs. a generic transformer); active learning; cost-sensitive evaluation (Wk 7).
What good looks like	A triage model with a precision/recall trade-off chosen for the PV use case, error analysis, and a human-in-the-loop workflow diagram.
Traps mentors will probe	Recall matters more than precision here — say so; regulatory reporting timelines; leakage from templated synthetic text.

11. MLR pre-review assistant

Track: GenAI · Course weeks: 7

Commercial question	Does a draft promotional claim match the label and the approved reference deck, and what would a reviewer flag?
Decision owner / behavior modeled	Medical-legal-regulatory review · reviewer judgement

Data plan	A public FDA label for a real (non-course) drug; a set of synthetic claims (some compliant, some not) you author with labels.
Methods	Retrieval-augmented generation over the label; claim-to-evidence matching; an evaluation harness with precision/recall on flagged claims and a false-negative analysis.
What good looks like	A tool that cites the exact label sentence supporting or contradicting each claim, with measured accuracy and an explicit “not a substitute for MLR” boundary.
Traps mentors will probe	Hallucinated citations; over-trust; failing on paraphrase; not measuring the cases that matter most (unsupported superiority claims).

12. Weekly brand-performance briefing agent

Track: Agentic AI (syllabus archetype) • Course weeks: 1, 3, 6, 10

Commercial question	Can an agent produce the Monday brand briefing — pull the data, detect what changed, draft the narrative and flag actions — reliably enough that an analyst only reviews it?
Decision owner / behavior modeled	Brand analytics team • analyst workflow
Data plan	Synthetic weekly tables: TRx/NRx/NBRx by segment and geography, share, access status, promotion activity, with injected anomalies.
Methods	Tool-using agent (SQL/pandas tools), anomaly detection, templated + LLM narrative, evaluation against analyst-written briefings; guardrails and logging.
What good looks like	Ten weeks of generated briefings scored for factual accuracy, a catalog of failure modes, and a cost/time comparison with the manual process.
Traps mentors will probe	Numbers in the narrative that don't match the tables; no evaluation; “agent” that is really a fixed script (be honest about which parts are autonomous).

13. Payer-policy monitoring agent

Track: Agentic AI • Course weeks: 2, 6

Commercial question	Which formulary and policy documents changed this month for our brand and competitors, what changed, and how many covered lives are affected?
Decision owner / behavior modeled	Market access • payer decisions
Data plan	Public formulary PDFs and policy web pages (Medicare Part D formulary files are public); a synthetic covered-lives table.
Methods	Document parsing, diffing, retrieval and summarization; change classification; a human-review queue; exposure calculation.
What good looks like	A monthly change report with source links, measured accuracy on a labeled set of changes, and a covered-lives exposure estimate.
Traps mentors will probe	PDF parsing errors silently dropping tables; summarizing changes that didn't happen; scope creep — pick 3–5 payers.

14. Synthetic advisory board

Track: GenAI • Course weeks: 7

Commercial question	Can LLM personas grounded in real interview transcripts or published physician surveys predict what an advisory board would say about a positioning question?
Decision owner / behavior modeled	Marketing and medical • HCP attitudes
Data plan	Public physician survey results and published qualitative studies; synthetic transcripts you construct; hold out a subset to test predictions.
Methods	Persona construction with retrieval grounding; structured elicitation; comparison of persona answers with held-out human answers; bias and consistency checks.

What good looks like	A measured agreement rate between synthetic and human responses, an analysis of where personas fail, and a recommendation on when the method is (not) usable.
Traps mentors will probe	Sycophantic personas; circularity (testing on what you trained on); overclaiming — IQVIA's own white paper frames this as a complement to, not a replacement for, real ad boards.

15. Forecast-explanation agent

Track: Agentic AI • Course weeks: 6, 10

Commercial question	When the forecast and actuals diverge, can an agent explain the variance to a brand lead in plain language — drivers, magnitude, confidence — and propose what to check?
Decision owner / behavior modeled	Brand and finance • forecast consumers
Data plan	Synthetic forecast vs. actuals with known injected drivers (access loss, competitor launch, inventory, seasonality).
Methods	Time-series decomposition and driver attribution; a tool-using agent that queries components and writes the explanation; evaluation against the injected ground truth.
What good looks like	Explanations that recover the true injected drivers most of the time, with a confusion analysis of the misses and a UI mock of how a brand lead would consume it.
Traps mentors will probe	Attributing variance to whichever driver is easiest to describe; ignoring interaction effects; no ground truth.

Proposal template (due to your mentor by Week 4, Oct 5)

Section	What to write (½ page total)
Problem statement	One paragraph: the commercial question, the decision owner, and why it matters (in \$ or in a stakeholder's words).
Stakeholder behavior modeled	HCP, payer, patient, or an internal process — and which data captures that behavior.
Data plan	What you will generate, license, or download; size; known limitations.
Methods and milestones	Baseline by check-in #1 (Nov 23); full model by check-in #2 (Dec 7); final Dec 14.
Evaluation	How you will know it works: metric, holdout, comparison to a naive baseline.
Risks	What could make the project fail and the fallback.